HANDS-ON 1 · DAY 1 / MORNING · PAIR'S SHEET

Watching the threshold, then choosing it

The case. A bank grants consumer loans. Its data science team has 1,000 past applicants in `german_credit.csv`: 22 columns describing each applicant, and a column `Score` saying whether the loan was repaid (`good`, 700) or not (`bad`, 300). The risk committee wants to see what a model would decide before it approves one.

STEP 1 · TRAIN A FIRST MODEL

1 ☐ **Load the data**
In your project: **+ Dataset** → **Upload your files**, choose `german_credit.csv`, create the dataset.

2 ☐ **Create the analysis**
Select the dataset, then **Lab** → **AutoML prediction**. Target `Score`. Keep **Quick prototypes**, create.

3 ☐ **Drop the identifier**
Design → **Features handling**: set `row ID` to **Reject**. An identifier describes nobody.

4 ☐ **Train**
Keep the default split and algorithms. Click **Train**, wait for the models to finish.

STEP 2 · NAME THE POSITIVE CLASS, THEN READ THE MATRIX

1 ☐ **Open the best model**
Results, click the model with the highest score, then **Performance** → **Confusion matrix**.

Which value of `Score` does Dataiku treat as **positive**? Read it from the row and column labels of the confusion matrix.

_____

At the threshold Dataiku shows by default (_____), record the four cells:

	predicted bad	**predicted good**
actually bad	TP _____	FN _____
actually good	FP _____	TN _____

Precision = TP / (TP + FP) = _____ Recall = TP / (TP + FN) = _____

STEP 3 · MOVE THE THRESHOLD

Drag the **threshold** slider above the confusion matrix and record what the same screen shows:

threshold	**precision**	**recall**	**good applicants refused (FP)**	**bad applicants granted (FN)**
0.3				
0.5				
0.7				

Lowering the threshold refuses more applicants. In the table, which of the two errors grows, and which shrinks?

☐ Reached this line? Stop here and wait for the costs.

The rest is for block 2, after the slides on costs, ROC and cross-validation.

4 · ENTER WHAT EACH ERROR COSTS

The committee fixes the costs: granting a loan to a **bad** applicant costs 5, refusing a **good** applicant costs 1, a right decision costs 0.

1 ☐ **Set the cost matrix**
Design → **Metrics**: under threshold optimization, optimize for **Cost matrix gain**. Enter the costs as negative gains: bad predicted good → −5, good predicted bad → −1, both right decisions → 0. Train again.

Threshold Dataiku now selects (**Performance** → **Confusion matrix**): _____ The rule from the slides gives 1 / (1 + 5) ≈ 0.17. Why might the two differ?

5 · REPLACE ONE SPLIT BY FIVE FOLDS

1 ☐ **Switch to cross-validation**
Design → **Train/Test set**: **K-fold cross-test**, 5 folds, stratified. Keep two algorithms: Logistic Regression and Random Forest. Train.

algorithm	ROC AUC	±
Logistic Regression		
Random Forest		

Does the ± let you say one algorithm is better? What would you need to see?

6 · READ THE GAINS CHART

On the best model: **Performance** → **Lift charts**. If analysts can only review the 20% of applicants the model finds riskiest, what share of the bad applicants will they see? _____